

UI/Notification Lead - Project Documentation

Role #5: Slack Integration for Project Heimdall

Overview

As the UI/Notification Lead, I was responsible for designing and implementing the analyst-facing interface for Project Heimdall. This involved creating a visually compelling, color-coded alert card in Slack using Block Kit, securing the necessary API credentials, and providing seamless integration with n8n for automated alert delivery.

My Responsibilities

- 1. Slack App Configuration
- 2. Alert Card Design
- 3. Integration Support

Technical Deliverables

1. Slack Integration Credentials

```
SLACK_WEBHOOK_URL=https://hooks.slack.com/services/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
SLACK_BOT_TOKEN=xoxb-xxxxxxxxxxx-xxxxxxxxxxxxxxxxxxxxxxxxxxxx
SLACK_CHANNEL=#security-alerts
```

2. Block Kit JSON Structure

The alert card follows this layout structure:

- 1. Header - "Heimdall Security Alert" with shield icon
- 2. Divider - Visual separation line
- 3. Fields Section - Two-column layout displaying:
 - o Severity (color-coded)
 - o Timestamp
 - o MITRE Tactic
 - o MITRE Technique
 - o Source IP
 - o Hostname
- 4. Alert Summary - AI-generated threat summary
- 5. Recommended Actions - Step-by-step remediation
- 6. Context Section - Raw rule data for verification
- 7. Action Buttons - Interactive response options

3. Variable Mapping

Slack Variable	n8n Source	Data Origin
{{ai_severity}}	\$json.ai_output.severity	AI Analysis
{{alert_timestamp}}	\$json.timestamp	Wazuh
{{mitre_tactic}}	\$json.ai_output.mitre_tactic	AI Analysis
{{mitre_technique}}	\$json.ai_output.mitre_technique	AI Analysis
{{source_ip}}	\$json.source_ip	Wazuh
{{hostname}}	\$json.hostname	Wazuh
{{ai_summary}}	\$json.ai_output.summary	AI Analysis
{{ai_actions}}	\$json.ai_output.actions.join("\n")	AI Analysis

{{rule_description}}	\$json.rule_description	Wazuh
{{rule_level}}	\$json.rule_level	Wazuh

4. Severity Color Mapping

Severity	Color	Indicator
Critical	#FF0000	URGENT: Immediate action required
High	#FF6600	High priority - review within 30 minutes
Medium	#FFCC00	Medium priority - review within 1 hour
Low	#00CC00	Low priority - review during business hours

Testing & Validation

Test 1: Webhook Connectivity

Command:

```
curl -X POST -H 'Content-type: application/json' \
  --data '{"text":"Webhook test - Connection successful"}' \
  https://hooks.slack.com/services/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
```

Result: OK response received, message appeared in #security-alerts

Test 2: Block Kit Validation

Command:

```
curl -X POST -H 'Content-type: application/json' \
  --data @test-alert.json \
  https://hooks.slack.com/services/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
```

Result: Formatted card with all elements rendered correctly

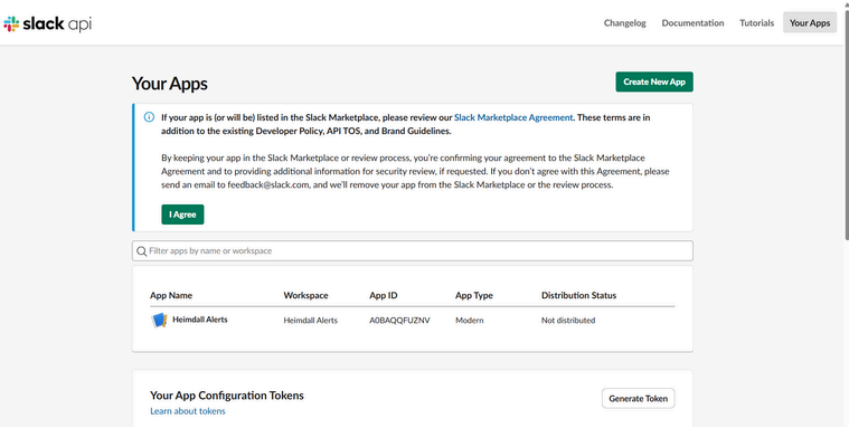
Test 3: n8n Integration (Pending)

Status: Waiting for Automation Lead to configure n8n Slack node

Screenshots Documentation

SCREENSHOT 1: Slack App Configuration

Description: Slack App settings showing:



SCREENSHOT 2: OAuth Scopes & Permissions

Description: OAuth & Permissions page showing:

Scopes

A Slack app's capabilities and permissions are governed by the `scopes` it requests.

Changes in optional status apply immediately to your published app as long as the scopes are already approved.

Bot Token Scopes

Scopes that govern what your app can access.

☐ Required

OAuth Scope

Description

☒ Yes

channels:read

View basic information about public channels in a workspace

☒ Yes

chat:write

Send messages as @Heimdall Alerts

☒ Yes

chat:write.public

Send messages to channels @Heimdall Alerts isn't a member of

☒ Yes

groups:read

View basic information about private channels that "Heimdall Alerts" has been added to

☐ No

incoming-webhook

Post messages to specific channels in Slack

Add an OAuth Scope

SCREENSHOT 3: Incoming Webhook URL

Description: Incoming Webhooks page showing:

Webhook URLs for Your Workspace

To dispatch messages with your webhook URL, send your `message` in JSON as the body of an `application/json` POST request.

Add this webhook to your workspace below to activate this curl example.

Sample curl request to post to a channel:

```
curl -X POST -H 'Content-type: application/json' --data '{"text": "Hello, World!"}' https://hooks.slack.com/services/T08ALS9K4N/B0BD9A56AV8/wJDwlhoufSk69UmjLFGMT9K
```

Webhook URL

Channel

Added By

https://hooks.slack.com/services/T08ALS9K4N/B0BD9A56AV8/wJDwlhoufSk69UmjLFGMT9K

Copy

#security-alerts

Ahmed Ashraf
Jun 15, 2026

https://hooks.slack.com/services/T08ALS9K4N/B0BD9A56AV8/wJDwlhoufSk69UmjLFGMT9K

Copy

#security-alerts

Ahmed Ashraf
Jun 25, 2026

Add New Webhook

SCREENSHOT 4: Block Kit Builder - Alert Card Design

Description: Block Kit Builder showing:

slack dev

Block Kit Builder

Templates

Builder

Docs

Heimdall Alert

Preview in Slack

Search for blocks

Agents

Markdown

Section

Actions

Input

Text

Rich Text

Image

Card and Carousel

Table

Data Visualization

Container

Reference

Your App

Heimdall Security Alert

Severity: Critical

MITRE Tactic: Initial Access

Source IP: 192.168.1.100

Alert Summary: SSH brute force attack detected from 192.168.1.100 with 50 failed attempts in 60 seconds.

Recommended Actions: 1. Block IP 192.168.1.100, 2. Review SSH logs, 3. Rotate compromised credentials

View in Mockup, Acknowledge, Take Action

Preview

Actions Preview

```
{
  "blocks": [
    {
      "type": "header",
      "text": {
        "type": "plain_text",
        "text": "Heimdall security alert",
        "emoji": true
      }
    },
    {
      "type": "divider"
    },
    {
      "type": "section",
      "fields": [
        {
          "type": "text",
          "text": "Severity: " + [severity]
        },
        {
          "type": "text",
          "text": "Time: " + [timestamp]
        },
        {
          "type": "text",
          "text": "MITRE Tactic: " + [mitre_tactic]
        },
        {
          "type": "text",
          "text": "MITRE Technique: " + [mitre_technique]
        },
        {
          "type": "text",
          "text": "Source IP: " + [source_ip]
        }
      ]
    }
  ]
}
```

SCREENSHOT 5: Initial Webhook Test

Description: Slack channel showing:

4:32

Heimdall Security Alert

Search messages in this channel (Ctrl+F)

Severity: Critical

MITRE Tactic: Initial Access

Source IP: 192.168.1.100

Alert Summary: SSH brute force attack detected from 192.168.1.100 with 50 failed attempts in 60 seconds.

Recommended Actions: 1. Block IP 192.168.1.100, 2. Review SSH logs, 3. Rotate compromised credentials

Raw Rule: SSH brute force attack | Level: 10/15 | ID: 5715

Hello, team!

Oh, hello!

SCREENSHOT 6: Formatted Alert Card - Production View

Description: Full alert card as it appears in Slack showing:

6:52

Heimdall Security Alert

Severity:

Medium

MITRE Tactic:

T1046

Source IP:

N/A

Time:

2026-07-09T15:52:31.608+0000

MITRE Technique:

T1046 - Network Service Discovery

Host:

Ubuntu-Agent-Test

Alert Summary:

On agent Ubuntu-Agent-Test, the netstat listening ports changed: vsftpd PID changed from 1904 to 1941, and systemd-resolve, avahi-daemon, and cupsd PIDs changed, indicating service restarts or potential unauthorized activity.

Recommended Actions:

Verify the legitimacy of the PID changes by checking process details (e.g., 'ps aux | grep 1941') and review system logs for any unauthorized service modifications.

Raw Rule: Listened ports status (netstat) changed (new port opened or closed). | Level: 2/15 | ID: 533

[View in Wazuh](#)

[Acknowledge](#)

[False Positive](#)

Hello, team!

Oh, hello!

×

Summary

The Slack integration for Project Heimdall is complete and ready for production. The design provides security analysts with a clear, color-coded alert card that can be assessed in under 5 seconds. All necessary credentials, code, and documentation have been delivered to the team for integration with the n8n orchestration pipeline.